

DOXYLIN WATER SOLUBLE POWDER 500MG/G

Ingredient(s):
Each g contains :
Doxycycline (as Doxycycline Hydrochloride)..... 500mg

Pharmacology (Summary of Pharmacodynamics and Pharmacokinetics):
Doxycycline has its main mechanism of action on protein synthesis. It is more lipid soluble than other Tetracyclines and passes directly through the lipid bilayer of the bacterial cell wall. In addition, an energy-dependent active transport system pumps the drug through the inner cytoplasmic membrane. Once inside the bacterial cell, Doxycycline inhibits protein synthesis by binding specifically to the 30S ribosomes. The drug appears to prevent access of aminoacyl tRNA to the acceptor site on the mRNA ribosome complex. This prevents the addition of amino acids to the growing peptide chain.
Doxycycline is well absorbed after oral administration. The drug is also readily absorbed in monogastric animals. Doxycycline absorption may only be reduced by 20% by either food or dairy products in the gut, and are excreted more slowly, leading to persistent serum levels. Doxycycline may be more extensively biotransformed (up to 40% of a given dose).
Plasma protein binding for Doxycycline is dependent upon species, and is proportional to the relative lipid solubility of the congener. Protein binding is 75~86% in dogs, and about 93% in cattle and pigs. Cats have higher binding than dogs.
Doxycycline is distributed widely to most body tissues, including heart, kidney, lungs, muscle, pleural fluid, bronchial secretions, sputum, bile, saliva, synovial fluid, ascitic fluid, and aqueous and vitreous humour. Doxycycline has the advantage of greater tissue penetration because of increased lipid solubility and an expended antimicrobial spectrum. Also, Doxycycline penetrates body tissues and fluids better than tetracycline HCl or Oxytetracycline, including to the CSF, prostate and eye
Doxycycline achieves greater tissue and tissue fluid concentrations in the prostate gland, saliva and tears, lungs and bronchial secretions. This drug is primarily excreted into the feces via non-biliary routes in an inactive form. In the dog, urinary, fecal, and biliary routes constitute 25%, 75%, and less than 5% of Doxycycline elimination. Biliary excretion accounts for a minor fraction of the drug elimination. Small intestinal excretion constitutes the major route of elimination. The drug is partially inactivated in the intestine by chelate formation and then excreted into the intestinal lumen. An inverse relationship exists between lipid solubility and urinary excretion, presumably because of greater tubular reabsorption of Doxycycline. The serum half-life of Doxycycline in dogs is approximately 10~12 hours. In calves, the drug has similar pharmacokinetic values. Doxycycline does not accumulate in patients with renal dysfunction.

Indication(s):
Treatment of chronic respiratory disease (CRD) and mycoplasmosis caused by microorganisms sensitive to doxycycline.

Dosage and Administration:
In drinking water use.
Chickens (broilers): 20 mg of doxycycline (equivalent to 40 mg of the product) / kg BW / day for 3 - 5 days.
mg product/L water = [40mg of product/kg b.w./day * mean body weight of the animal to be treated (kg)] / water consumed by kg b.w (l).
For the preparation of the medicated water the body weight of the animals to be treated and their actual daily water intake should be taken into due account. To provide the required amount of product in mg per litre drinking water the following calculation should be made:

=
$$\frac{40 \text{ mg product /kg bodyweight / day} \times \text{Average body weight (kg) animals to be treated}}{\text{Average amount of drinking water / animal (l)}}$$

= X mg product /l of drinking water
To ensure a correct dosage body weight should be determined as accurately as possible to avoid underdosing.
The intake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of doxycycline has to be adjusted accordingly. Do not use at concentrations lower than 0.23 g of powder /l in drinking water with pH higher or equal to 7.5 to avoid precipitation.
Sufficient access to the system of water supply should be available for the animals to be treated to ensure adequate water consumption. No other source of drinking water should be available during the medication period.
Only sufficient medicated drinking water should be prepared to cover daily requirements. Orally via drinking water.

Target Species
Chickens (broilers).

Contraindications:
1. Contraindicated in patients hypersensitive to doxycycline.
2. Until further studies documenting the safety of intravenous doxycycline in horses are done, the parenteral route of administering this drug in horses should be considered contraindicated.

Precaution(s) / Warning(s):
1) Oral doxycycline has been implicated in causing esophageal strictures in cats.
2) Tetracycline therapy (especially long-term) may result in overgrowth (superinfections) of non-susceptible bacteria or fungi.

Interaction with Other Medicaments:
1. Most interactions are based on a physicochemical reaction in the gastrointestinal tract which reduces the bioavailability of the Tetracyclines. The most common interacting substances are the divalent or trivalent cations present in antacids, saline cathartics, milk and other foods, and calcium, magnesium, aluminium, zinc or bismuth cations preparations. The administration of Doxycycline should not be followed by the above substance in less than 2~3 hours.
2. Barbiturate drugs and carbamazepine decrease the serum half-life of Doxycycline.
3. Oral sodium bicarbonate, kaolin, pectin or bismuth subsalicylate, and laxatives reduce GI absorption of Doxycycline when given together orally.
4. Microsomal enzyme inducers, such as phenobarbital and phenytoin, shorten the plasma half-life of Doxycycline.
5. Riboflavine reduced the antimicrobial activity in vitro of Doxycycline HCl by 56.5%.
6. Oral iron products decreased Tetracycline absorption. Concurrent oral ferrous sulphate administration lowered the serum concentration of Doxycycline administered by mouth. Administration of iron salts should preferable be given 3 hours before or 2 hours after the Tetracycline dose.
7. Bacteriostatic drugs like the Tetracyclines, may interfere with bactericidal activity of the Penicillins, Cephalosporins, and Aminoglycosides.
8. Tetracyclines may depress plasma prothrombin activity and patients on anticoagulant (e.g., Warfarin) therapy may need dosage adjustment.
9. GI side effects may be increased if Tetracyclines are administered concurrently with Theophylline products.

Pregnancy and Lactation:
Because tetracyclines can retard fetal skeletal development and discolor deciduous teeth, they should only be used in the last half of pregnancy when the benefits outweigh the fetal risks.

Side Effect(s) / Adverse Reaction(s):
1. Mild yellow ultra-violet fluorescence was noted in the liver, kidneys and bones and small amounts of intracytoplasmic granular material were present in thyroid tissue.
2. The most common side effects of oral Tetracyclines use in small animals is vomiting and diarrhea.
3. Cause permanent staining of developing teeth and hypoplasia of enamel.
4. Staphylococcal enterocolitis due to resistant organisms or pseudo-membranous colitis due to infection with *Clostridium difficile* may also occur.
5. Overgrowth of resistant organisms may occur, causing glossitis, stomatitis and vaginitis, as well as Staphylococcal enterocolitis.
6. Bulging fontanelles may occur in young and benign intracranial hypertension in adults.
7. The antianabolic effect of Tetracyclines may cause an increase in blood urea.

Environmental Property:
None has been mentioned in animal.

Symptoms and Treatment for Overdosage, and Antidote(s):
Oral administration of divalent or trivalent cation antacids may bind some of the drug and reduce GI distress.

Shelf-Life:
4 years.
Reconstituted preparations should be used up within 24 hours.

Storage Condition(s):
Store at temperature below 30°C.

Product Description and Packing(s):
Yellow color fine powder and without other foreign matter.
Al. Pouch of 500g and 1kg

For food producing animals product:
Withdrawal Periods
Chicken meat & offal = 6 days
Egg: Do not use in laying hen from which eggs are produced for human consumption.

Maximum Residual Limit (MRL):
Poultry: Muscle 100µg/kg
Skin & fat 300µg/kg
Liver 300µg/kg
Kidney 600µg/kg

Disposal:
Dispose of any unused product or containers in accordance with guidance of the local waste regulation authority.



Made in Vietnam
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